Out of Boundary Paths

1. Aim

To write a C program to calculate the number of paths by which a ball can move out of the boundary of a grid within a given maximum number of moves.

2. Theory

The Out of Boundary Paths problem involves an \(m \times n\) grid and a ball initially placed at a specified cell. In each move, the ball can move up, down, left, or right by one cell.

The objective is to find the number of ways the ball can move outside the grid boundary in at most a specified number of moves.

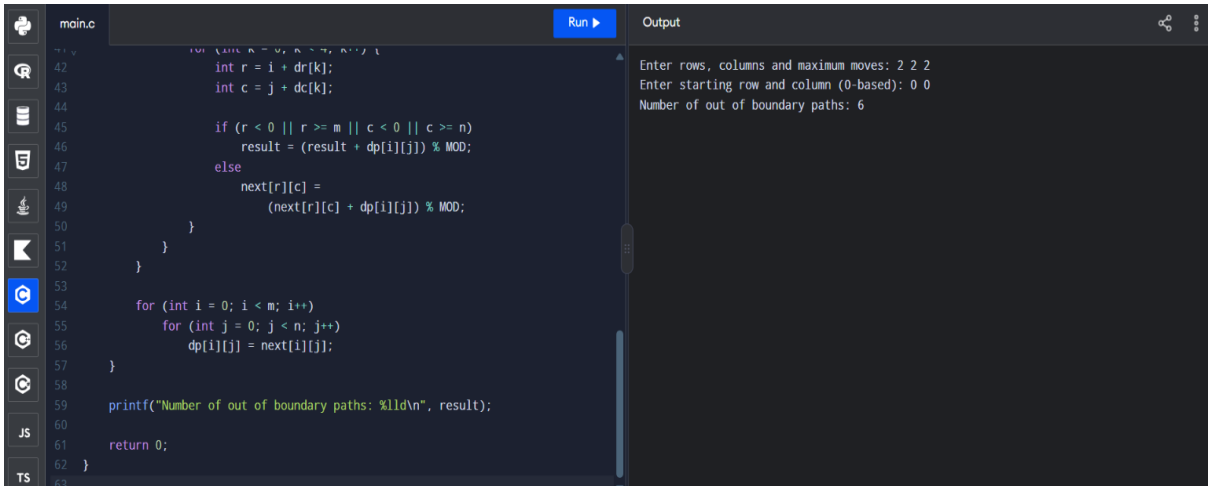
Dynamic Programming is used to solve this problem efficiently. The algorithm tracks the number of ways to reach each cell after a given number of moves. Whenever a move takes the ball outside the grid, that path is counted as a successful path.

The answer is calculated modulo \(10^9+7\) to prevent large numbers from exceeding the supported integer range.

3. Algorithm

1. Start.
2. Read the grid dimensions, maximum number of moves, and starting position.
3. Initialize the number of ways to reach the starting cell to 1.
4. For each move, examine every cell in the grid.
5. Distribute the number of ways to its four neighboring cells.
6. If a move goes outside the grid, add those ways to the answer.
7. Repeat until the maximum number of moves is reached.
8. Display the total number of paths.

9. Stop.



7. Complexity Analysis

Let \(r\) be the queried row.

- Time complexity: \(O(r^2)\), because the algorithm processes each glass in all rows up to row \(r\).
- Space complexity: \(O(r^2)\) for storing the tower.

8. Result

The amount of champagne present in the specified glass was calculated successfully using Dynamic Programming.